Curriculum Vitae

Ziyan WU

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Education

PhD supervised by **Prof. Rui Tang** and **Prof. Ivan Korolija** Institute for Environmental Design and Engineering, The Bartlett, University College London, United Kingdom	Nov 2022 – Nov 2026 (expected)
M.Sc. in Advanced Materials Science Institute for Materials Discovery, Faculty of Mathematical and Physical Sciences, University College London, United Kingdom	Sep 2020 – Sep 2021
B.Eng. (Hons) in Mechanical Engineering Faculty of Science and Engineering, University of Nottingham, Ningbo, China	Sep 2016 – Sep 2020

Research Interests

My research focuses on **intelligent building control** and **building-grid interactions**, with an emphasis on multi-building coordinated control for demand response, voltage regulation, and demand flexibility provision. I work on developing advanced and practical **Reinforcement Learning** algorithms, particularly safe RL, for modern building energy systems and power systems.

In parallel, I work on constructing co-simulation framework of simulation programs (e.g., EnergyPlus and Modelica) to build scalable and physics-based, open-source **control testbeds** that support RL implementations, control analysis and benchmarking for smart building/grid applications.

Publications

MuFlex: A Scalable, Physics-based Platform for Multi-Building Flexibility Analysis and Coordination
Energy (JCR Q1, IF 9.4), 2026.
Wu, Z., Korolija, I., and Tang, R.

Reinforcement learning in building controls: A comparative study of algorithms considering model availability and policy representation
Journal of Building Engineering (JCR Q1, IF 7.4), 2024.
Wu, Z., Zhang, W., Tang, R., Wang, H., and Korolija, I.

Assessing the Impacts of Energy Sharing on Low Voltage Distribution Networks: Insights into Electrification and Electricity Pricing in Germany
Applied Energy (JCR Q1, IF 11.0), in collaboration with University of Bayreuth, 2025.
Lersch, J., Tang, R., Weibelzahl, M., Weissflog, J., and **Wu, Z.**

Grid-aware Multi-building Coordination of Voltage Regulation using Constrained Reinforcement Learning
Arxiv, under review, 2026.
Wu, Z., Korolija, I., and Tang, R.

Safe Reinforcement Learning for Building Control: A Review and Standardized Benchmark Suite
Manuscript preparation, 2026.
Wu, Z., Korolija, I., and Tang, R.

Impact of Model Discrepancy on Reinforcement Learning Performance in Building HVAC Control
Manuscript preparation, in collaboration with University of Toronto, 2026.
Wu, Z., et al.

Conference Papers and Presentations

CIBSE Technical Symposium **March 2026**

A High-fidelity Multi-Building Simulation Platform for Reinforcement Learning-based Coordinated Control Strategies Training and Benchmarking. **Wu, Z.**, Tang, R.

Building Simulation **June 2026**

Multi-Building Coordinated Control of Grid Voltage on a Physics-Based Simulation Platform Using Constrained Reinforcement Learning. **Wu, Z.**, Tang, R. (**Best Paper Award**)

Project Experiences

Review and Benchmark Suite on Safe Reinforcement Learning for Building Control **Apr 2026**

Developed a standardized benchmark suite for safe RL for building controls, supporting single-/multi-building, DERs, and power grid co-simulation and integrating advanced safe RL algorithms. Multi-scenario benchmarking and literature review have been completed, and manuscript is currently in preparation.

Constrained RL for Grid Voltage Regulation **March 2026**

Developed a novel constrained RL algorithm, CSAC-VR, for coordinating multi-building HVAC and PV inverters to improve thermal comfort and energy efficiency while ensuring voltage safety. The paper was shortlisted for the Best Paper Award and is currently under review at *Building Simulation 2026*.

MuFlex **Feb 2026**

Developed and open-sourced *MuFlex*, a scalable, physics-based platform for multi-building flexibility analysis and coordination. The platform was validated on a multi-building demand-limiting task using SAC algorithm, and supports scalability tests of 50 buildings. The work was published in *Energy*.

Investigating Building Model Discrepancy impact on RL Performance **Nov 2025**

Collaborated with Prof. Seungjae Lee's group at the University of Toronto to study how discrepancies in pre-trained building models affect the test-time control performance of RL agents. The experiments are near completion, and the manuscript is being finalized.

Energy sharing in Germany LV networks **Jan 2025**

Collaborated with Prof. Martin Weibelzahl's group at the University of Bayreuth to study the impacts of energy sharing on German low-voltage distribution networks under different DER penetration levels, pricing schemes, and electrification scenarios. The work was published in *Applied Energy*.

Benchmarking Advanced RL algorithms for Building Control **May 2024**

Conducted a systematic benchmark of RL algorithms for building control, comparing value-based, policy-gradient, actor-critic, and model-based methods in terms of control performance, data efficiency, and hyperparameter robustness. The work was published in the *Journal of Building Engineering*.

Teaching Experiences

Teaching Assistant, 2022-2026

Machine Learning in Smart Buildings (MSc module), University College London

Python Tutor, 2022-2026

Smart Buildings and Digital Engineering (MSc program), University College London

Teaching Assistant, 2019

Manufacturing Automation, University of Nottingham, Ningbo, China

Awards

Dean's Scholarship, University of Nottingham Ningbo China, 2017